Guilherme Monteiro

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AI engineer who is also product- and client-oriented: I lead client interaction, design the product, and build it end to end, from Figma and frontend to backend, architecture, AI models and hardware installation. At ALTAVE, a safety-critical monitoring company, I was Forward Deployed Engineer for the NABORS drilling client, manage the SATORO maritime project, and led the architecture of a next-generation real-time GPU vision pipeline (DeepStream, TensorRT). I ship models end to end, deployed LLM-based alert validation, and run a multi-agent engineering workflow daily. Mechatronics Engineering, Poli-USP.

Core Competencies

- **Real-Time GPU Inference**: DeepStream, TensorRT, ONNX, CUDA; multi-stream video decode, batched inference, detection delivery to downstream consumers.
- **Computer Vision**: PyTorch, TensorFlow, Keras, YOLOv8 / YOLOv8-seg, FastSAM, OpenCV; image classification, instance segmentation, object detection, multi-model systems, synthetic data generation.
- **Inference Architecture**: concurrency models, service boundaries, configurable inference layers with minimal shared contracts, hexagonal architecture, incremental migrations, architecture decision records.
- **Python Services and LLMs**: advanced Python, FastAPI, gRPC, asyncio; LLM-based validation of safety-equipment alerts in production.
- **Platform and Tooling**: Linux and Git daily, YouTrack, Docker, AWS, MLflow, SQL, Power BI, Power Apps, Power Automate.
- **Reinforcement Learning (independent)**: self-play, MCTS, learned policy/value networks.

Professional Experience

- **ALTAVE** — **São José dos Campos, Brazil**
  Applied ML Engineer, Computer Vision & Platform — *Jun 2024 - Present*
  Senior IC on the AI team, also owning product roadmap and client interface. ALTAVE builds safety-critical monitoring technology; I work across client delivery, product design and engineering.

  NABORS (drilling): Forward Deployed Engineer, Jun - Aug and Oct - Nov 2026 (5 months on site abroad)
  - Led client interaction with the client's operations teams and owned the delivery end to end: product design, development, architecture, AI and hardware installation.
  - Product design: designed an after-action review product for rig operations from requirements to Figma, then built its frontend and backend.
  - Architecture: drove the target architecture for the vision platform (site-scoped rules, an edge runtime that closes the loop with no cloud link, and a cloud control plane), with 43 platform requirements traced to design elements.
  - AI: fine-tuned detection models for personal-protective-equipment monitoring on annotated drill-floor data and wired alert emission into the platform.
  - Hardware: installed and commissioned camera, recorder and edge-box hardware at the rig and mapped camera channels into the platform.

  SATORO (maritime): Project Manager, since joining ALTAVE (Jun 2024)
  - Ran the project roadmap and delivery for a vessel-based monitoring line: onboard detection, a bridge application for the crew, and shore-side sandbox and data services.
  - Delivered person and vessel detection running live inference on onboard video, and traced and fixed the model-format and GPU-memory defects that blocked it.
  - Drove bridge application work: camera views, footage access with scoped login, and vessel-tracking overlays.

  Other projects

- **Blowout preventer (BOP) landing monitoring**: contributed to development of a photogrammetry and depth-inference system that gives a bird's-eye view of subsea BOP landing operations in deep water.
- **Railway obstruction monitoring**: architected the solution and managed the developers. Analysed 67,000 alert thumbnails, showed that 21 of 119 camera scenes produced 92% of false alerts, and recommended a per-scene temporal model over the per-frame classifier.
- **Suspended load detection**: architected the solution and managed the developers; tuned motion-based detectors (background subtraction and optical flow) for detecting loads and rigging.

Core engineering

- Build real-time GPU vision pipelines on DeepStream and TensorRT: multi-stream video decode, batched inference, and detection delivery to downstream consumers.
- Led the architecture of the next-generation vision pipeline: concurrency model, service boundaries, and an incremental migration path for existing consumers, with tradeoffs recorded as architecture decision records.
- Consolidated a fragmented set of single-purpose detection services into one configurable inference layer behind a deliberately minimal shared contract.
- Delivered an end-to-end oil-spill crisis-management platform for offshore clients: instance segmentation for spill boundaries, object detection for vessels, people and fauna, and an operational coordination interface on AWS.
- Own the full model lifecycle for an on-vessel detector: synthetic dataset generation, YOLOv8-seg fine-tuning on a single consumer GPU, and ONNX export against the shipboard serving contract.
- Built a zone-evaluation library on hexagonal architecture, released alongside its predecessor so consumers could migrate without a coordinated cutover.
- Deployed LLM-based validation of safety-equipment alerts that cut false positives by over 95%, at the cost of one Gemini inference per alert.
- Onboarded 10+ engineers in a high-turnover, high-growth environment, sustaining velocity through written architecture decisions.
- Build typed, contract-first Python services on AsyncAPI and JSON Schema contracts with generated Pydantic v2 models.
- Legolas, a lean edge camera runtime (successor to Sauron): measured about 31 MB of extra memory per additional FHD camera against a 250 MB budget, with a design target of at least 9 concurrent FHD cameras per x86_64 host.
- Run a multi-agent engineering workflow daily (leads, workers, supervision and decision routing) to deliver across these projects.
- **Stack:** Python, PyTorch, TensorRT, DeepStream, ONNX, YOLOv8, FastSAM, OpenCV, CUDA, asyncio, gRPC, FastAPI, TypeScript, Figma, Docker, AWS, MLflow.

Braskem — **São Paulo, Brazil**
Product Analyst — *2023 - 2024*

- Analysed operating indicators and ticket trends for the company's HCM product and prioritised system fixes, cutting ticket volume by 10% over the year.
- Built applications and automations for process governance in the People area, consolidating information exchange and senior-management approvals.
- Ran governance and process knowledge for the HCM system: global trainings and consolidated documentation.

Independent project — **Remote**
Freelance Backend Engineer — *2026 (4 months)*

- Built the whole backend of an electoral-analytics platform: a FastAPI service over public electoral and census data in BigQuery, a PostgreSQL store with Alembic migrations, JWT authentication, rate limiting, transactional email, Docker and GCP deployment, and a test suite (about 175 commits).

Digital Social — **São Paulo, Brazil**
Business Intelligence Analyst (freelance) — *Apr 2023 - Oct 2023*

- Built KPI reports showing clients' return on investment and automated spreadsheets and indicators for the internal consultants.

- **Braskem** — **São Paulo, Brazil**
  People Analytics Intern — *Aug 2022 - Jun 2023*
  - Automated the HCM data distribution flow, from a week of manual work to immediate.
  - Built access-control applications for HCM data that ensured LGPD compliance, and attrition dashboards with sentiment analysis of free-text answers.
- **Kinea Investimentos** — **São Paulo, Brazil**
  Commercial Department — *May 2021 - Apr 2022*
  - Investor contact and liaison with relationship teams at other market firms through fund-status update meetings and recurring support materials.
  - Managed the company's contact structure in Salesforce CRM (accounts, contacts, reports, campaigns); competitor, market and economic studies.

Independent Projects

- **vision-agent-mcp** (github.com/guilherme-r-monteiro/vision-agent-mcp): an MCP server that gives an LLM agent five computer-vision tools (YOLOv8n on ONNX Runtime), with a 36-case evaluation harness that scores the agent against COCO128 labels. A one-sentence tool-description change that makes the agent retry at a lower confidence lifted it from 26/36 to 28/36, against a 26/36 detector-only baseline. Typed Python, 14 unit tests, results and limits published.
- **Facial-expression effort model** (2022-2023): convolutional network with transfer learning predicting effort level in physical activity from facial expressions; data augmentation and cleaning; over 95% accuracy on low data with no overfitting.
- **HelperBot** (github.com/guilherme-r-monteiro/helper-bot): AlphaZero-style self-play agent (MCTS plus a learned policy/value network) playing KeyForge end to end through the open-source keyteki engine; stochastic, large action space, no clean tree to search. Hexagonal layers across Python/PyTorch training and Node.js/ONNX inference; spec-driven, with 13 architecture decision records, a PRD and the full task tree committed alongside the code.

Education

- **Escola Politécnica da USP** — **São Paulo, Brazil**
  Bachelor's in Mechatronics Engineering — *Mar 2018 - Jul 2023*
  Mechatronics, Robotics, Control and Automation Engineering. Tutor, Programa de Tutoria da Poli (Mar 2021 - Mar 2022); Physics Monitor, Cursinho da Poli (Mar 2020 - Dec 2020).

Certifications

- CPA-20 (2021).

Languages

- Portuguese (native), English (native or bilingual).

References

- Available upon request.